

DATA SHEET

TDA4662 Baseband delay line

Product specification
Supersedes data of 1995 Oct 30
File under Integrated Circuits, IC02

1996 Nov 14

Baseband delay line

TDA4662

FEATURES

- Two comb filters, using the switched-capacitor technique, for one line delay time (64 μ s)
- For PAL and NTSC
- Adjustment-free application
- Handles negative or positive colour-difference input signals
- Clamping of AC-coupled input signals [$\pm(R-Y)$ and $\pm(B-Y)$]
- VCO without external components
- 3 MHz internal clock signal derived from a 6 MHz CCO, line-locked by the sandcastle pulse (64 μ s line)
- Sample-and-hold circuits and low-pass filters to suppress the 3 MHz clock signal
- Addition of delayed and non-delayed output signals
- Output buffer amplifiers
- Comb filtering functions for NTSC colour-difference signals to suppress cross-colour.

GENERAL DESCRIPTION

The TDA4662 is an integrated baseband delay line circuit with one line delay. It is suitable for PAL and NTSC decoders with colour-difference signal outputs $\pm(R-Y)$ and $\pm(B-Y)$.

QUICK REFERENCE DATA

SYMBOL	PARAMETER	MIN.	TYP.	MAX.	UNIT
V_{P1}	analog supply voltage (pin 9)	4.5	5	6	V
V_{P2}	digital supply voltage (pin 1)	4.5	5	6	V
$I_{P(tot)}$	total supply current	–	5.5	7.0	mA
$V_{i(p-p)}$	$\pm(R-Y)$ input signal PAL/NTSC (peak-to-peak value; pin 16)	–	525	–	mV
	$\pm(B-Y)$ input signal PAL/NTSC (peak-to-peak value; pin 14)	–	665	–	mV
G_v	voltage gain V_O/V_I of colour-difference output signals				
	V_{11}/V_{16} for PAL and NTSC	5.3	5.8	6.3	dB
	V_{12}/V_{14} for PAL and NTSC	5.3	5.8	6.3	dB

ORDERING INFORMATION

TYPE NUMBER	PACKAGE		
	NAME	DESCRIPTION	VERSION
TDA4662	DIP16	plastic dual in-line package; 16 leads (300 mil)	SOT38-4
TDA4662T	SO16	plastic small outline package; 16 leads; body width 3.9 mm	SOT109-1

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BLOCK DIAGRAM

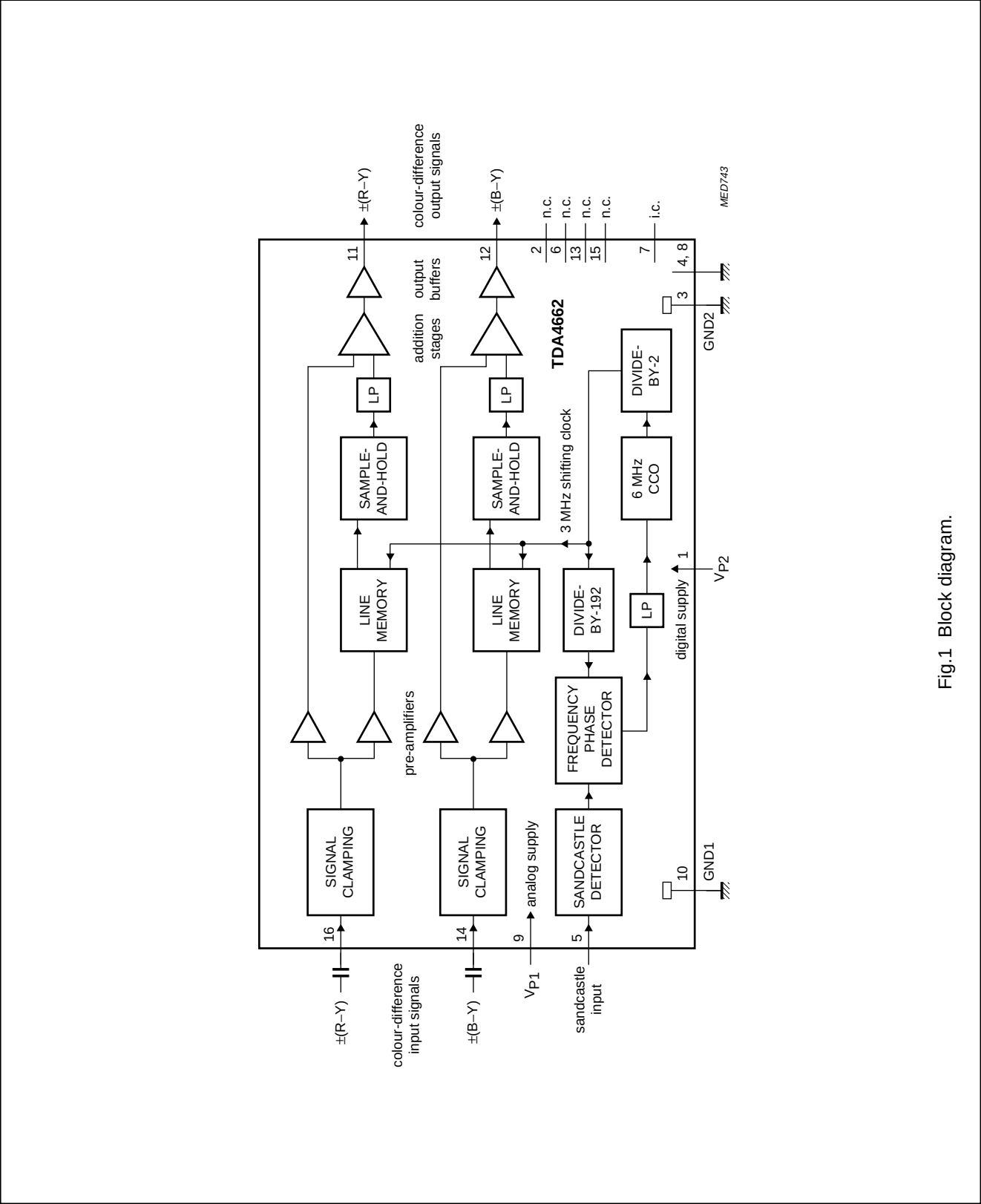


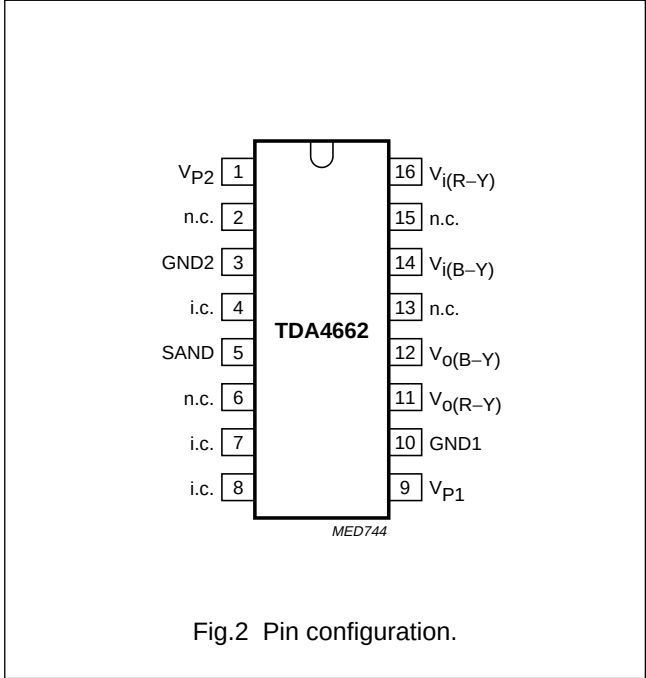
Fig.1 Block diagram.

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PINNING

SYMBOL	PIN	DESCRIPTION
V _{P2}	1	supply voltage for digital part (+5 V)
n.c.	2	not connected
GND2	3	ground for digital part (0 V)
i.c.	4	internally connected
SAND	5	sandcastle pulse input
n.c.	6	not connected
i.c.	7	internally connected
i.c.	8	internally connected
V _{P1}	9	supply voltage for analog part (+5 V)
GND1	10	ground for analog part (0 V)
V _{o(R-Y)}	11	±(R-Y) output signal
V _{o(B-Y)}	12	±(B-Y) output signal
n.c.	13	not connected
V _{i(B-Y)}	14	±(B-Y) input signal
n.c.	15	not connected
V _{i(R-Y)}	16	±(R-Y) input signal



LIMITING VALUES

In accordance with the Absolute Maximum Rating System (IEC 134). Ground pins 3 and 10 connected together.

SYMBOL	PARAMETER	CONDITIONS	MIN.	MAX.	UNIT
V _{P1}	supply voltage (pin 9)		-0.5	+7	V
V _{P2}	supply voltage (pin 1)		-0.5	+7	V
V ₅	input voltage on pin 5		-0.5	V _P + 1.0	V
V _n	voltage on pins 11, 12, 14 and 16		-0.5	V _P	V
T _{stg}	storage temperature		-25	+150	°C
T _{amb}	operating ambient temperature		0	70	°C
V _{ESD}	electrostatic handling for all pins	note 1	-	±500	V

Note

1. Equivalent to discharging a 200 pF capacitor through a 0 Ω series resistor.

THERMAL CHARACTERISTICS

SYMBOL	PARAMETER	VALUE	UNIT
R _{th j-a}	thermal resistance from junction to ambient in free air		
	SOT38-4	75	K/W
	SOT109-1	220	K/W

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CHARACTERISTICS

$V_P = 5\text{ V}$; input signals as specified in characteristics with 75% colour bars; super-sandcastle frequency of 15.625 kHz; $T_{\text{amb}} = 25\text{ }^{\circ}\text{C}$; measurements taken in Fig.3; unless otherwise specified.

SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
V_{P1}	analog supply voltage (pin 9)		4.5	5	6	V
V_{P2}	digital supply voltage (pin 1)		4.5	5	6	V
I_{P1}	analog supply current (pin 9)		–	4.8	6.0	mA
I_{P2}	digital supply current (pin 1)		–	0.7	1.0	mA
Colour-difference input signals						
$V_{i(p-p)}$	input signal (peak-to-peak value) $\pm(R-Y)$ PAL and NTSC (pin 16) $\pm(B-Y)$ PAL and NTSC (pin 14)		–	525	–	mV
			–	665	–	mV
$V_{i(\text{max}; p-p)}$	maximum symmetrical input signal (peak-to-peak value) $\pm(R-Y)$ for PAL and NTSC $\pm(B-Y)$ for PAL and NTSC	before clipping	660	–	–	mV
		before clipping	840	–	–	mV
$R_{14,16}$	input resistance during clamping		–	–	40	k Ω
$C_{14,16}$	input capacitance		–	–	10	pF
$V_{14,16}$	input clamping voltage	proportional to V_P	1.3	1.5	1.7	V
Colour-difference output signals						
$V_{o(p-p)}$	output signal (peak-to-peak value) $\pm(R-Y)$ on pin 11 $\pm(B-Y)$ on pin 12		–	1.05	–	V
			–	1.33	–	V
V_{11}/V_{12}	ratio of output amplitudes at equal input signals	$V_{i14,16} = 665\text{ mV (p-p)}$	–0.4	0	+0.4	dB
$V_{11,12}$	DC output voltage	proportional to V_P	2.5	2.9	3.3	V
$R_{11,12}$	output resistance		–	330	400	Ω
G_v	gain for PAL and NTSC	ratio V_o/V_i	5.3	5.8	6.3	dB
$V_{N(\text{rms})}$	noise voltage (RMS value; pins 11 and 12)	$V_{i14,16} = 0\text{ V}$; note 1	–	–	1.2	mV
$V_{11,12(p-p)}$	unwanted signals (line-locked) (peak-to-peak value) meander spikes	$V_{i14,16} = 0\text{ V}$; active video; $R_S = 300\text{ }\Omega$	–	–	5	mV
			–	–	10	mV
$S/N(W)$	weighted signal-to-noise ratio (pins 11 and 12)	$V_o = 1\text{ V (p-p)}$; note 1	–	54	–	dB
t_d	time difference between undelayed and delayed output signals (pins 11 and 12)		63.94	64	64.06	μs
	delay of undelayed signals		40	60	80	ns

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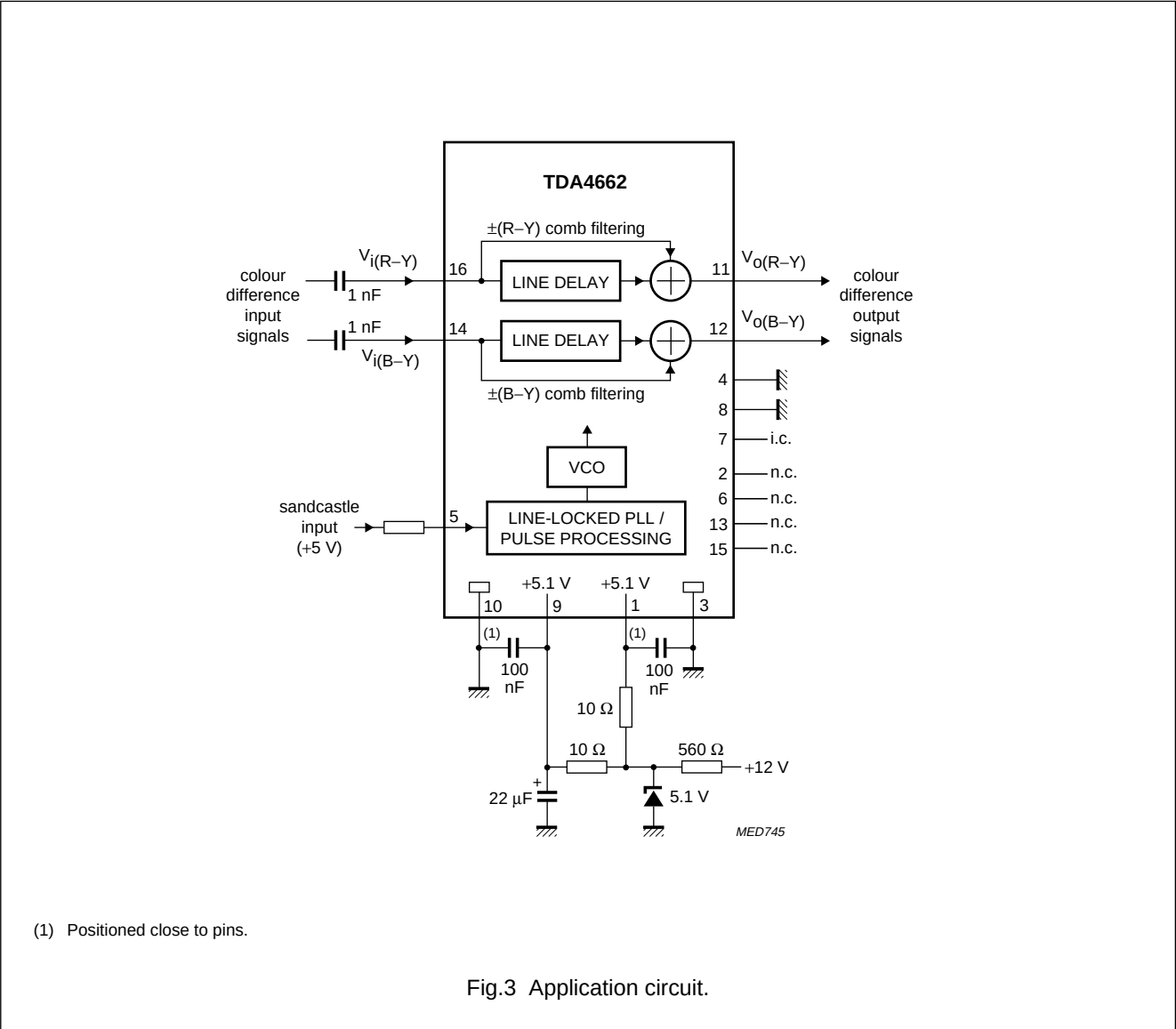
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SYMBOL	PARAMETER	CONDITIONS	MIN.	TYP.	MAX.	UNIT
Sandcastle pulse input (pin 5)						
f _{BK}	burst-key frequency/sandcastle frequency		14.2	15.625	17.0	kHz
V ₅	top pulse voltage	note 2	4.0	–	V _P + 1.0	V
V _{slice}	internal slicing level		V ₅ – 1.0	–	V ₅ – 0.5	V
I ₅	input current		–	–	10	µA
C ₅	input capacitance		–	–	10	pF

Notes

- 1. Noise voltage at f = 10 kHz to 1 MHz; R_S < 300 Ω.
- 2. The leading edge of the burst-key pulse or H-blanking pulse is used for timing.

APPLICATION INFORMATION



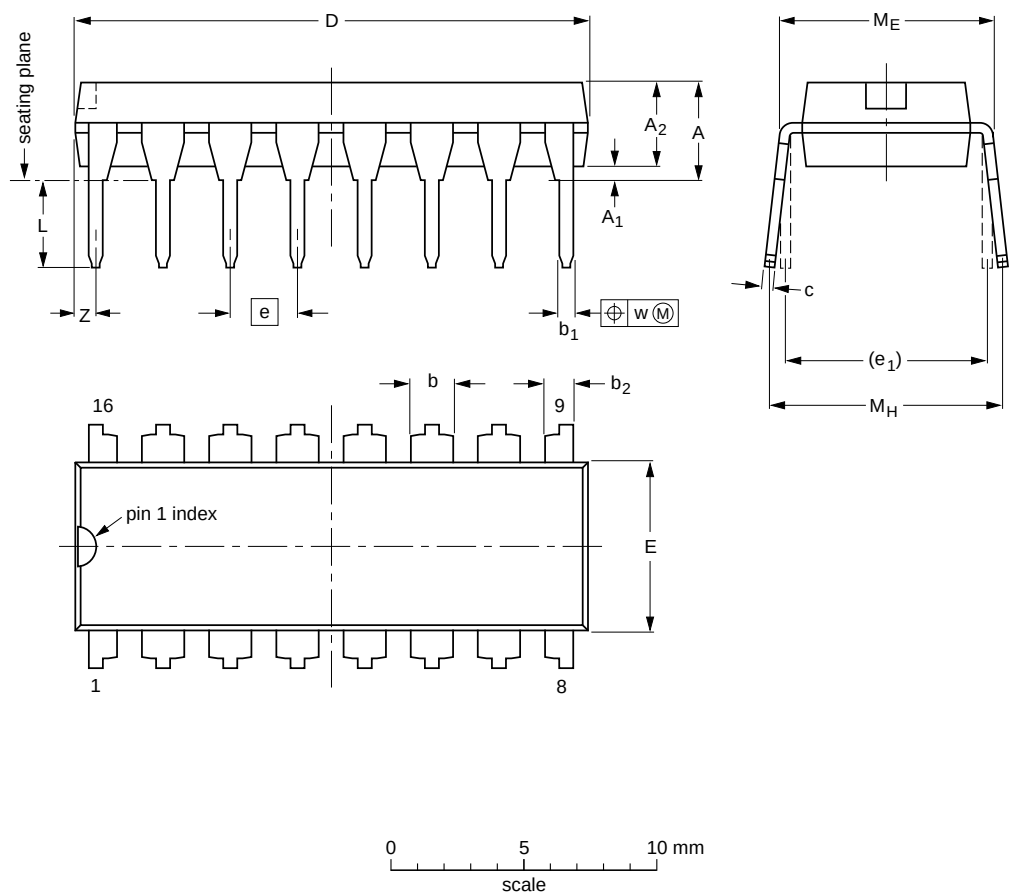
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PACKAGE OUTLINES

DIP16: plastic dual in-line package; 16 leads (300 mil)

SOT38-4



DIMENSIONS (inch dimensions are derived from the original mm dimensions)

UNIT	A max.	A ₁ min.	A ₂ max.	b	b ₁	b ₂	c	D ⁽¹⁾	E ⁽¹⁾	e	e ₁	L	M _E	M _H	w	Z ⁽¹⁾ max.
mm	4.2	0.51	3.2	1.73 1.30	0.53 0.38	1.25 0.85	0.36 0.23	19.50 18.55	6.48 6.20	2.54	7.62	3.60 3.05	8.25 7.80	10.0 8.3	0.254	0.76
inches	0.17	0.020	0.13	0.068 0.051	0.021 0.015	0.049 0.033	0.014 0.009	0.77 0.73	0.26 0.24	0.10	0.30	0.14 0.12	0.32 0.31	0.39 0.33	0.01	0.030

Note

1. Plastic or metal protrusions of 0.25 mm maximum per side are not included.

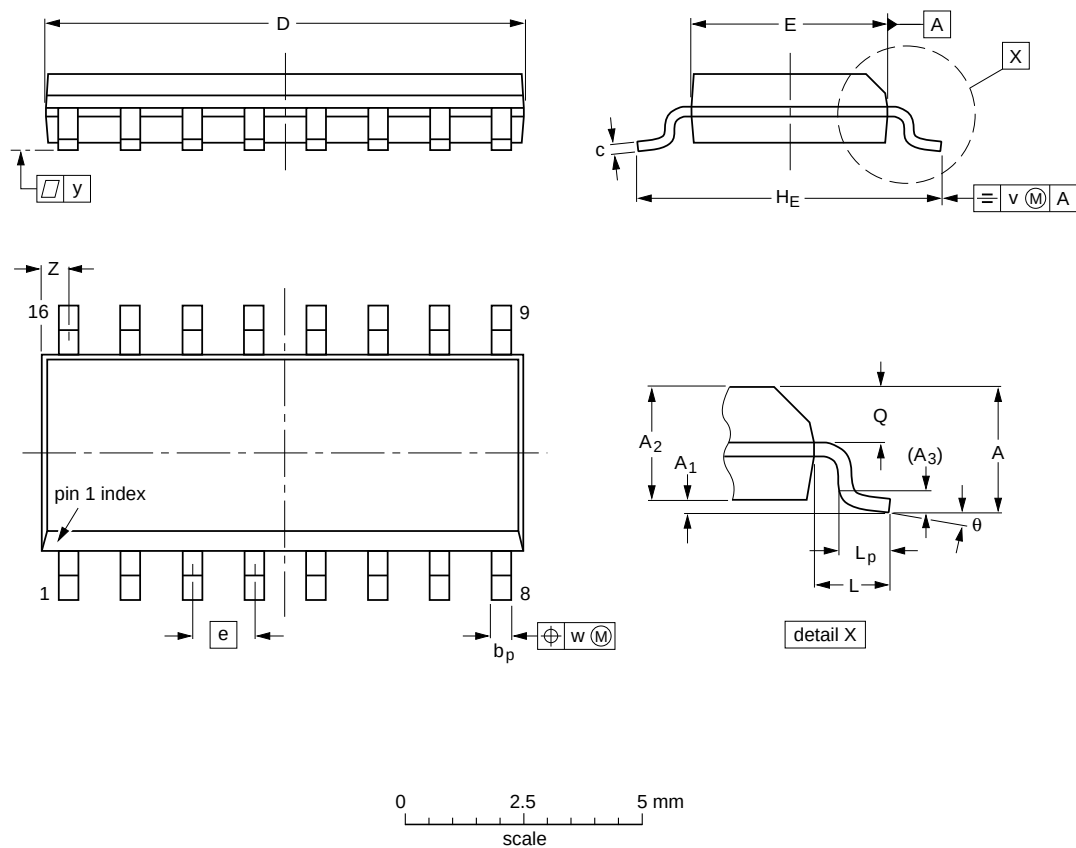
OUTLINE VERSION	REFERENCES				EUROPEAN PROJECTION	ISSUE DATE
	IEC	JEDEC	EIAJ			
SOT38-4						92-11-17 95-01-14

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SO16: plastic small outline package; 16 leads; body width 3.9 mm

SOT109-1



DIMENSIONS (inch dimensions are derived from the original mm dimensions)

UNIT	A max.	A ₁	A ₂	A ₃	b _p	c	D ⁽¹⁾	E ⁽¹⁾	e	H _E	L	L _p	Q	v	w	y	Z ⁽¹⁾	θ
mm	1.75	0.25 0.10	1.45 1.25	0.25	0.49 0.36	0.25 0.19	10.0 9.8	4.0 3.8	1.27	6.2 5.8	1.05	1.0 0.4	0.7 0.6	0.25	0.25	0.1	0.7 0.3	8° 0°
inches	0.069	0.0098 0.0039	0.057 0.049	0.01	0.019 0.014	0.0098 0.0075	0.39 0.38	0.16 0.15	0.050	0.24 0.23	0.041	0.039 0.016	0.028 0.020	0.01	0.01	0.004	0.028 0.012	

Note

1. Plastic or metal protrusions of 0.15 mm maximum per side are not included.

OUTLINE VERSION	REFERENCES				EUROPEAN PROJECTION	ISSUE DATE
	IEC	JEDEC	EIAJ			
SOT109-1	076E07S	MS-012AC				91-08-13 95-01-23

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SOLDERING

Introduction

There is no soldering method that is ideal for all IC packages. Wave soldering is often preferred when through-hole and surface mounted components are mixed on one printed-circuit board. However, wave soldering is not always suitable for surface mounted ICs, or for printed-circuits with high population densities. In these situations reflow soldering is often used.

This text gives a very brief insight to a complex technology. A more in-depth account of soldering ICs can be found in our *"IC Package Databook"* (order code 9398 652 90011).

DIP

SOLDERING BY DIPPING OR BY WAVE

The maximum permissible temperature of the solder is 260 °C; solder at this temperature must not be in contact with the joint for more than 5 seconds. The total contact time of successive solder waves must not exceed 5 seconds.

The device may be mounted up to the seating plane, but the temperature of the plastic body must not exceed the specified maximum storage temperature ($T_{stg\ max}$). If the printed-circuit board has been pre-heated, forced cooling may be necessary immediately after soldering to keep the temperature within the permissible limit.

REPAIRING SOLDERED JOINTS

Apply a low voltage soldering iron (less than 24 V) to the lead(s) of the package, below the seating plane or not more than 2 mm above it. If the temperature of the soldering iron bit is less than 300 °C it may remain in contact for up to 10 seconds. If the bit temperature is between 300 and 400 °C, contact may be up to 5 seconds.

SO

REFLOW SOLDERING

Reflow soldering techniques are suitable for all SO packages.

Reflow soldering requires solder paste (a suspension of fine solder particles, flux and binding agent) to be applied to the printed-circuit board by screen printing, stencilling or pressure-syringe dispensing before package placement.

Several techniques exist for reflowing; for example, thermal conduction by heated belt. Dwell times vary between 50 and 300 seconds depending on heating method. Typical reflow temperatures range from 215 to 250 °C.

Preheating is necessary to dry the paste and evaporate the binding agent. Preheating duration: 45 minutes at 45 °C.

WAVE SOLDERING

Wave soldering techniques can be used for all SO packages if the following conditions are observed:

- A double-wave (a turbulent wave with high upward pressure followed by a smooth laminar wave) soldering technique should be used.
- The longitudinal axis of the package footprint must be parallel to the solder flow.
- The package footprint must incorporate solder thieves at the downstream end.

During placement and before soldering, the package must be fixed with a droplet of adhesive. The adhesive can be applied by screen printing, pin transfer or syringe dispensing. The package can be soldered after the adhesive is cured.

Maximum permissible solder temperature is 260 °C, and maximum duration of package immersion in solder is 10 seconds, if cooled to less than 150 °C within 6 seconds. Typical dwell time is 4 seconds at 250 °C.

A mildly-activated flux will eliminate the need for removal of corrosive residues in most applications.

REPAIRING SOLDERED JOINTS

Fix the component by first soldering two diagonally-opposite end leads. Use only a low voltage soldering iron (less than 24 V) applied to the flat part of the lead. Contact time must be limited to 10 seconds at up to 300 °C. When using a dedicated tool, all other leads can be soldered in one operation within 2 to 5 seconds between 270 and 320 °C.

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DEFINITIONS

Data sheet status	
Objective specification	This data sheet contains target or goal specifications for product development.
Preliminary specification	This data sheet contains preliminary data; supplementary data may be published later.
Product specification	This data sheet contains final product specifications.
Limiting values	
Limiting values given are in accordance with the Absolute Maximum Rating System (IEC 134). Stress above one or more of the limiting values may cause permanent damage to the device. These are stress ratings only and operation of the device at these or at any other conditions above those given in the Characteristics sections of the specification is not implied. Exposure to limiting values for extended periods may affect device reliability.	
Application information	
Where application information is given, it is advisory and does not form part of the specification.	

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